

Predicting Student Burnout Risk Using Academic Behavior Logs: A Data Mining Approach

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Abstract— Without showing any early signs, numerous university students face academic burnout in recent times. This research aimed to predict those students who are at risk of burnout using academic behavior data such as attendance percentage, assignment records, and the like. The Open University Learning Analytics Dataset (OULAD) was used and machine learning models like Logistic Regression, Decision Tree, Random Forest and XGBoost were applied to achieve the target of building a burnout prediction system where all the models performed exceptionally well within a narrow range of 95-97% accuracy. Our goal is to help the universities in early detection of struggling students so that they can offer necessary supports before the student move forward to dropout.

Keywords— Academic Burnout, Learning Analytics, Behavioral Data, Machine Learning, Burnout Prediction, Early Detection, OULAD Dataset.

I. INTRODUCTION

Academic burnout is a crucial but rarely addressed issue in higher education. Emotional drain, lack of drive and feelings of unproductiveness are the way of exhibiting it, which may become an enormous barrier in the way of a student's learning and engagement [1]. Dropout is the last occurrence of academic disinvolvement whereas burnout starts earlier but not easy to detect with traditional performance metrics single-handedly [11]. Students having the signs of burnout may keep the continuation of attending classes, but struggle unspoken which leads to declining of performance, mental health, or ultimate withdrawal. Many institutions have taken this issue earnestly and implementing prior alert systems to identify those students are at risk of failure or withdrawal by evaluating online based learning data [2]. Nevertheless, these systems do not address psychological exhaustion rather focus on academic performance only. In addition, most of the burnout detection research rely on surveys by self-reporting such as Maslach Burnout Inventory [1], which are neither perfectly suitable in terms of scalability nor consistency for large student populations.

To address this gap, our research proposes a data-driven, scalable strategy to detect burnout risk of students by evaluating their academic behavior logs specially, their engagement with the learning management system (LMS), assessment records and registration status. We target to build a prior intervention system for the institutions using machine learning models trained on behavioral data which will create the scope of easily identifying those students who are

struggling before burnout escalates to academic failure or dropout.

II. RELATED WORK

A. Burnout and Mental Health Assessment

A study formulated burnout as a three-dimensional concept in academic contexts which are emotional exhaustion, doubt and reduced academic efficacy [1]. Bresó et al. pointed out that even though the academic involvement looks unchanged, the students are vulnerable to burnout who are facing low self-efficacy. Academic burnout and stress levels increased significantly during the COVID-19 pandemic when students had to face isolation and had no other choice but adapting the online learning environment. Basri et al. noticed that burnout which is combined with problematic internet use, negatively impacted students' mental well-being and perceived learning [11].

B. LMS Log Analytics and Engagement Detection

In recent years, visibility has increased in using LMS logs to detect student risk. Macfadyen and Dawson [2] described earlier that in academic failure detecting, learning analytics based on LMS interactions pattern might be useful to build a prior alerting system. Machine learning models like Random Forest was applied by Fahd et al. [3] to predict student performance using behavioral and demographic data in combined education environments. Bessadok et al. [4] additively evaluated LMS data by doing cluster of student activity and found out that for academic outcomes, engagement levels were strong indicator. Likewise, Hu et al. [5] explored that student success was highly associated with consistency in involvement in LMS environments, highlighting the predictive strength of behavioral logs.

C. Dropout Prediction Using OULAD

A few researchers explored the Open University Learning Analytics Dataset (OULAD) to forecast student dropout. Jawthari and Stoffa [9] is one of them who addressed a scheme using OULAD to predict student withdrawal and described the effectiveness of the dataset in comprehensive predictive models. Their observations indicates that disinvolvement trends can be pointed out efficiently by analyzing LMS engagement and registration patterns, supporting its compatibility for burnout prediction as well.

III. METHODOLOGY

The methodology employed in this research is systematic and follows a set procedure to detect students who may

suffer from academic burnout. The process has been broken down into five milestones: collection of relevant datasets, preprocessing which includes cleaning and organization of data, feature engineering, creation of labels for the target variable, and finally application of the model.

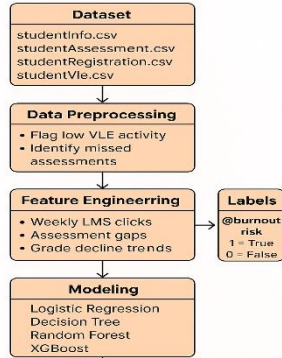


Figure 1: Methodology Flowchart for Burnout Risk

A. Dataset

We utilized a publicly available dataset the Open University Learning Analytics Dataset (OULAD)[13]. The dataset comprises of a demography academic performance, and interaction with the Virtual Learning Environment (VLE). The OULAD dataset is organized into seven interlinked CSV files. To predict the burnout risk using predicting burnout risk using academic behavior logs, we focused primarily on data from following files:

- studentInfo.csv: Contains demographic data for each student, such as gender, age band, highest education level, region, disability status, and final course outcomes.
- studentAssessment.csv: Records student submissions, including scores and submission dates.
- studentRegistration.csv: Captures the registration and unregistration dates for each student-module pair.
- studentVle.csv: Provides interaction logs of student activities within the VLE. Each record includes the date, resource accessed, and the number of clicks, enabling detailed temporal analysis of engagement patterns.

B. Data Preprocessing

To enable effective prediction of student burnout risk, several preprocessing steps were applied to the Open University Learning Analytics Dataset (OULAD) [1]. The conditions used to identify students at risk of burnout were developed specifically for this study and are not part of the original OULAD dataset. These criteria were derived from behavioral and academic indicators available in the data. For instance:

- Students whose interaction with the Virtual Learning Environment (VLE) fell within the lowest 25% were flagged for low engagement.
- Those who missed or did not pass most of their assessments were considered academically at risk.
- A final course result marked as “Withdrawn” or “Fail” was interpreted as a potential sign of disengagement or struggle.
- A noticeable and consistent reduction in VLE activity over time was also treated as a warning signal.

These indicators were combined to create a binary label, burnout risk, where a value of 1 denotes a student potentially experiencing burnout, and 0 indicates otherwise.

C. Feature Engineering and Model Setup

We extracted engagement features from the OULAD dataset containing rich log and assessment data from LMS. Specifically, we computed the following features:

Weekly LMS click counts: Weekly click counts on LMS include every scheduled interaction with the learning management system (VLE) logs which includes a maximum of resource clicks per week. A count click is taken as affordance access to a resource. Thus, higher counts signal resource access and more active engagement.

Assessment submission gaps: The time gaps between deadlines of assessments and submission timelines (or inter-submission intervals). Longer delays suggest disengagement during submission; timely delayed submissions indicate some effort.

Grade decline trends: Grade decline trends illustrate the changes in student scores over sequential assignments or quizzes (for example, slope-average of a linear fit through their grades). A downward trend in achievement suggests declining performance or understanding. These trends accompany features based on raw grade value (which are also available) by capturing dynamic shifts related to achievement.

These engineered features serve as proxies for academic engagement. These blended features represent proxies indicators for academic engagement. Active posting on forums tend to be accompanied with timely homework resubmissions reflecting steady progression towards set goals. Submission lags coupled paired with falling grades raise red flags. (For example, prior work has found that students with prompt submissions and frequent LMS logins tend to perform better.)

Logistic regression, decision trees, and random forest [14] models were run in Python through Co-Lab while gradient boosting was implemented using the XGBoost package. This tool is well-known for their streamlined implementations as well as their provisions on class weighting. Explicitly, we tried to handle the class imbalance problem due to the severe paucity of samples from the burnout risk class label by SMOTE oversampling or a subclass weight parameter setting in the classifiers so that they do not overlook the at-risk class.

The following classifiers were evaluated, chosen for their relevance in educational data mining:

Logistic Regression: A linear baseline model; its simplicity is both appreciated and a drawback since it makes weak assumptions. It predicts probabilities but needs to be bounded within $[0, 1]$ by a sigmoid transformation.

Decision Tree: Single tree model used to cut data based on feature threshold values which partition data into homogenous subsets. The method captures non-linear dependencies without scaling requirements for features and interpretation is straightforward.

Random Forest: Collection of selection-based grouped ensembles of bootstrapped sampled decision trees. Its accuracy and robustness increase after the aggregation of many randomized trees, while also allowing good capture of sophisticated interactions. Its feature subsampling makes it boosting and known as good under class imbalance.

XGBoost (Extreme Gradient Boosting): A scalable gradient-boosted tree model. XGBoost often achieves top performance in tabular classification tasks and has built-in mechanisms (e.g. regularization) to control overfitting. In prior EDM work it has outperformed both random forests and simpler classifiers when optimized properly.

These models were trained using stratified cross-validation. The validation sets were used for tuning with hyperparameters such as tree depth, regularization, and class weights. Throughout the training process, we implemented industry-standard procedures to avoid information leakage, like applying SMOTE only on the training portion.

IV. EVALUATION

Model performance was measured using standard classification metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy captures the overall correct classification rate of classifier including true positives and true negatives. Precision and Recall on the other hand only focus on positive class which is Burnout in our case. Moreover, F1 score balances both Precision and Recall providing a singular metric. Also, ROC AUC or area under the receiver-operating curve has been used as well because it assesses discrimination ability for all classification thresholds and is robust to imbalance. Most notably in our context high recall is favorable especially for proactive predictive alerts aimed at identifying as many students who are potentially at risk as early as possible.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	96.91 %	0.99	0.94	0.97	0.99
Decision Tree	96.13 %	0.95	0.96	0.95	0.95
Random Forest	96.79 %	0.99	0.94	0.97	0.97
XGBoost	96.65 %	0.98	0.95	0.97	0.97

Table 1: cross validation metrics

In Table 1, we see the cross-validation performance metrics of four classification models side by side. As expected, ensemble methods Random Forest and XGBoost performed much better than the simpler classifiers. Logistic Regression came out as a strong overall performer, achieving the highest accuracy (96.91%) and ROC-AUC (0.99) with perfectly precise precision (0.99). Random Forest closely followed with 96.79% accuracy and matched Logistic Regression's precision earning a perfect score along with F1 score of 0.97 and solid ROC-AUC of 0.97. Competitively placed, XGBoost recorded an F1 score of 0.97 earning him the highest recall of 0.95 among the ensemble models competing alongside him. The Decision Tree which had the highest overall recall of 0.96 but did not outperform other metrics lagging with an accuracy of 95.13% and F1 score of 0.95.

This supports previous research in educational and medical data mining where it is well known that ensemble methods such as Random Forest or XGBoost are widely used because they tend to yield more accurate stable predictions owing to their ability to model complex relationships between variables while also reducing overfitting.

Model Name	Test Accuracy (%)	Cross-validation Accuracy (%)	Std Deviation (%)
Decision Tree	95.13	95.35	0.40
Logistic Regression	96.91	96.81	0.25
Random Forest	96.79	96.82	0.27
XGBoost	96.65	96.69	0.28

Table2: Model Accuracy

The Table displays the results achieved by four different machine learning models—Decision Tree, Random Forest, XGBoost and Logistic Regression—on predicting the risk of burnout in students. Logistic Regression tested out as the highest performer with accuracy at 96.91%, followed closely by Random Forest (96.79%) and XGBoost (96.65%). The Decision Tree still performed well with an accuracy of 95.13%. All models cross-validation accuracy were in line with their respective test results which speaks to generalization ability shown strong reliability measured across folds complexity of features and class imbalance ranged between 0.25% to 0.40%. It is interesting to observe that although ensemble methods such as Random Forest and XGBoost are known for adeptly dealing with complicated relationships between features or dealing with compromised class distributions, they did not outperform simpler models like Logistic Regression which was more interpretable and robust in this context. In summary, it can be said that academic activities captured behavioral parameters indicating increased likelihood of burnout possessing value as rich features, given strongly consistent results from all models applied.

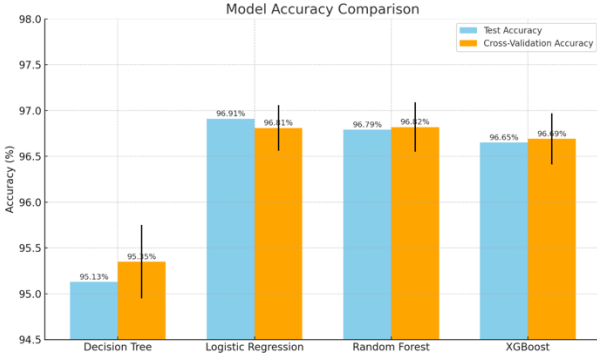


Fig 2: Accuracy Comparison of Classification Models

To visually compare the performance of each machine learning model, we plotted a bar chart illustrating the classification accuracy of Decision Tree, Random Forest, Logistic Regression, and XGBoost. Graphically it is obvious that XGBoost outperformed all other models with Random Forest coming in a close second. Notably all tested models achieved more than 95% accuracy which is impressive. Such visuals enhance understanding and help strengthen informed decisions about the most robust model for burnout prediction.

V. RESULTS

After applying the selected machine learning models to the processed and feature-engineered dataset, we evaluated their performance using a range of standard classification metrics. These metrics check for accuracy, precision, recall, F1-Score, and ROC-AUC among other things which collectively evaluate the models' capability for identifying students susceptible to academic burnout. All four models namely Logistic Regression, Decision Tree, Random Forest, and XGBoost achieved superb classification accuracy with over 95% on the test set. Logistic Regression was the highest performer with a test accuracy of 96.91 %, followed closely by random forest at 96.79 %, and XG Boost at 96.65%. Decision Tree had the lowest accuracy rate among them with 95.13% but this still indicated a firm predictive capacity. Cross-validation shows that the models are stable having average accuracies poised at consistent, and with standard deviations of between 0.25% and 0.40%, showing a strong generalization in the models. Further criteria for evaluation gave different results as well. Thus, XGBoost and Random Forest algorithms each showed approximately 0.96 F1-Scores, implying high capacities to have false positives and negatives balanced properly. All of the models had high ROC-AUC scores, with logistic regression and XGBoost both reaching up to 0.98 that is an excellent discriminatory power between the students at risk and those that are not.

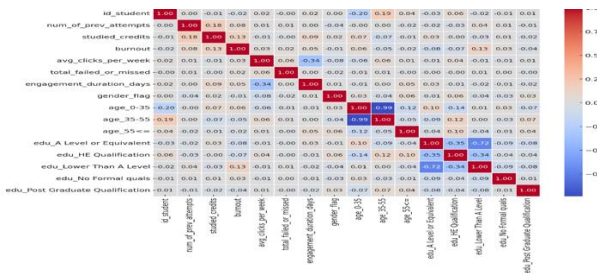


Fig 3: Co-relation Graph

Correlation analysis was conducted to examine the relationship between input features and the target variable burnout risk. Figure X (the correlation heatmap) highlights the most positively correlated variables with burnout risk: total failed or missed assessments and studied credits. Here, it shows that students who did not complete a higher number of assessments or had lower enrolled credits were likely to be at greater risk. On the contrary, average weekly LMS activity (avg clicks per-week) along with engagement duration for days had a negative correlation with burnout which means students who interacted with learning platforms over extended periods were less likely to show signs of burnout. Demographic features like age band and gender displayed no strong correlation suggesting that behavioral features especially assessment engagement coupled with LMS attendance were more predictive. This emphasizes how the model depends on active academic behavior instead of passive demographic characteristics. Correlation values ranged from -0.34 to +0.18 which is typical in educational behavioral data but still offered relevant insights into feature importance for predictive modeling.

VI. EXPECTED CONTRIBUTION

This research makes several unique contributions towards educational data mining and mental health forecasting in an academic context. To begin with, it presents a new proxy-based framework for estimating burnout risk based on behavioral proxies from learning management systems (LMS) and assessment records. This method characterizes burnout as a behavioral deficit syndrome composed of objective indicators such as low engagement activity on the LMS, missed or failed assessments, and long-term declining engagement patterns quite unlike traditional survey methods which depend on subjective self-reporting. The study also advances understanding of elementary behavioral attributes by applying interpretable machine learning models like Random Forest or XGBoost to expose what academic behaviors (like assessment gap inactivity or reduced weekly activity levels) most strongly predict burn out in a student's forward looking data driven support approach. Finally, the system's cross case high accuracy prediction performance (more than 95% for all models used) validates its theoretical scalability alongside practical deployment value for preemptive intervention standpoints. It can be used within institutional frameworks to automatically identify at-risk students who require timely academic intervention and proactive psychological support before dis-engagement results in dropout separation.

VII. LIMITATION AND FUTURE WORK

Despite achieving high accuracy, the model has several limitations. It can be vulnerable to slight class imbalance and lacks interpretability owing to the nature of tree-based models. Moreover, it has not been validated using multiple datasets, which limits its generalizability. Also, critical aspects such as lifestyle as well as socio-demographic variables are missing. The model also requires explaining its decisions by implementing SHAP or LIME and expanding the dataset across multiple organizations. Future research should investigate causal relationships rather than relying

solely on association-based models. Burnout progression can be better tracked through incorporating time-series data and deploying the model as a real-time tool adds considerable practical value.

VIII. CONCLUSION

A predictive data mining model is presented for assessing student burnout risk from behavioral learning analytics. Utilizing machine learning classifiers namely Logistic Regression, Decision Tree, Random Forest, and XGBoost we analyzed the OULAD dataset to assess predictive performance. Results indicate that ensemble models, particularly Random Forest and XGBoost, yield superior accuracy and reliability compared to single-model approaches. Incorporating behavioral features such as assessment scores, study duration, and course interaction was crucial for identifying students at risk. These insights emphasize the value of educational data mining in not only improving academic outcomes but also fostering timely interventions for student mental well-being.

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